

Sakshi Sharma

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AI/ML engineer with hands-on experience in Machine Learning, Deep Learning, NLP, and Computer Vision, focused on building real-world AI solutions through innovation and practical implementation. Strong problem-solving mindset with experience working on research-driven, industry-oriented, and hackathon-based projects in emerging AI technologies.

EDUCATION

B. Tech in Data Science (2023–2027) IPS Academy IES, Indore CGPA 9+
12th Grade (2022) – Karnataka Vidya Niketan

WORK EXPERIENCE

THDCIL - Ministry of Power – September 2025 – Present

Working on backend and AI/ML pipelines for secure document verification. Implemented authenticity scoring, forgery detection, and hash-based tamper checks to automate verification workflows.

Autonomous Vehicle for Calamity Rescue (AVCR- Academic project got Funding) – An ML-powered system deployed on Jetson Nano to assess survivability in disaster-hit, confined spaces. Uses real-time sensor data and models like Isolation Forest, LSTM, and Random Forest for detection, prediction, and classification.

Data Analytics Virtual Experience, Deloitte Australia (Forage) – April 2025 Completed a simulation involving data analysis and forensic tech; built a Tableau dashboard and used Excel for data classification and business insights.

PROJECTS

CarbonVault — AI Carbon Credit Platform - Full-stack platform automating carbon credit issuance for NGOs, admins, and corporates — eliminating manual land verification. Integrated satellite imagery with NDVI analysis for automated land cover validation and role-based audit workflows. Built with FastAPI, React, SpacetimeDB, and PostgreSQL; deployed on Vercel with multi-user authentication.

Auto Mail Apply - Agentic AI pipeline that automates the entire job application process — from scraping listings to sending tailored emails with zero manual effort. Uses NLP to match and rank candidate profiles against job descriptions, then auto-composes outreach via Python NLP pipelines. Built with FastAPI and Selenium; designed to scale across multiple job platforms and candidate profiles.

Rockfall Prediction & Alert System – Developed an ML-based system to predict rockfall risks in open-pit mines using geological and vibration sensor data. Implemented risk-level classification and real-time alerting to enhance worker safety and prevent accidents.

AI-Powered Personal Legal Assistant – Web-based AI tool combining LLM-powered document summarization, legal drafting, case strategy advisory, and document classification in one interface. Uses RAG to ground responses in relevant legal context, reducing hallucination and improving accuracy. Built to support individuals and small firms in streamlining legal research and drafting workflows.

Pothole Detection – Built a computer vision solution using YOLO and OpenCV to detect potholes from street-level images. Aims to support smart infrastructure management and automated road safety monitoring.

ACHIEVEMENTS

Winner – Techno Maths Presentation

Winner – Science Exhibition 2024

Winner – UDAAN'26 Phase I and Phase II

Winner – VyaparX Idea Pitching Competition

Winner – Inter-College Chess Competition 2023, 25

Smart India Hackathon (SIH) Finalist | NEC (National Entrepreneurship Challenge) Finalist 2024'25 | Top 20- Clear Skies Innovation Challenge | Top 100 Campus Ambassador, IIT Bombay

Participations hackathons – Bhartiya Antariksha Hackathon, Hacksungum, GDSC Codespire, Pravah 3.0, Hackdata' 25, Build with India, Samsung Solve for Tomorrow, Flipkart Grid 6.0 (Robotics), Snow Hacks

SKILLS

Programming & Scripting Python, C, C++

Machine Learning & AI Machine Learning, Deep Learning (MLP, CNN, RNN), NLP, LSTM, LLM, Transformers, Hugging Face, LangChain, RAG, Prompt Engineering, Fine-tuning, Reinforcement Learning, Scikit-learn, TensorFlow, PyTorch, Keras

Computer Vision OpenCV, YOLOv8, Image Segmentation, MediaPipe

Data & Analytics SQL, MS Excel, Power BI, Big Data, Data Analysis, EDA, MySQL, PostgreSQL, ChromaDB (Vector DB), Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Web & Backend Django, FastAPI, HTML, CSS, Bootstrap, REST APIs, WebSockets

Cloud & MLOps Vercel, Docker, Google Colab, MLflow, AWS (familiar)

Tools & Platforms Git, GitHub, VS Code, Jetson Nano

Soft Skills Problem-solving, Leadership, Calm under pressure, Presentations, Agentic Workflow Design

CERTIFICATION

AWS Cloud Operations - Training Badge

RDBMS PostgreSQL

Python Tutorial (IIT Bombay)

Artificial Intelligent and machine learning (NPTEL)

LEADERSHIP

Founder – E-Cell, IES IPS Academy

Core Member – Bad Talk